

Basic Concepts of Quantum Mechanics

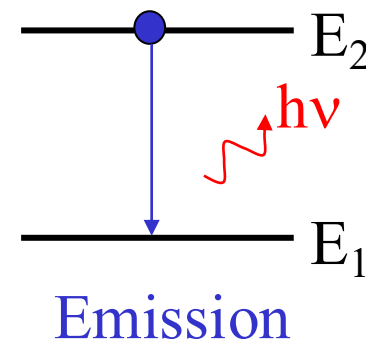
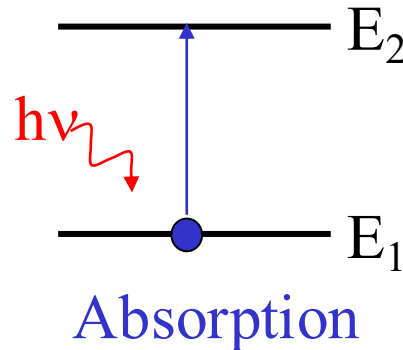
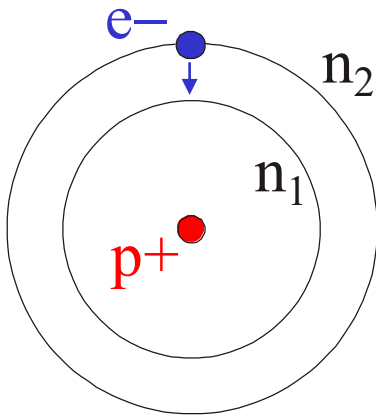
Classical (Newtonian) mechanics is inaccurate when applied to the systems with atomic dimensions.

Quantum mechanics is necessary background knowledge for understanding electrons in atoms.

The Bohr Model of the Atom

- The H-atom emission spectrum was rationalized by Bohr (1913):
 - Energies of H atom are restricted to certain discrete values (i.e. electron is restricted to well-defined circular **orbits**, labelled by **quantum number n**).
 - Energy (light) absorbed in discrete amounts (**quanta = photons**), corresponding to differences between these restricted values:

$$\Delta E = E_2 - E_1 = h\nu$$



Conclusion: Spectroscopy provides direct evidence for **quantization of energies** (electronic, vibrational, rotational etc.) of atoms and molecules

Limitations of Bohr Model & Rydberg-Ritz Equation

- The model only works for hydrogen (and other one electron ions) – **ignores e-e repulsion**.
- Does not explain **fine structure** of spectral lines.
- **Note:** The Bohr model (assuming circular electron orbits) is fundamentally incorrect.

Wave-Particle Duality

Light: wave nature (diffraction, refraction, interference....)
particle nature (photoelectric effect, Compton effect)
 ↓
 called “photon”

Electron: particle nature (m_0, q )
 wave nature (diffraction, refraction, interference...)

de Broglie's matter wave: All particles have the properties of wave.

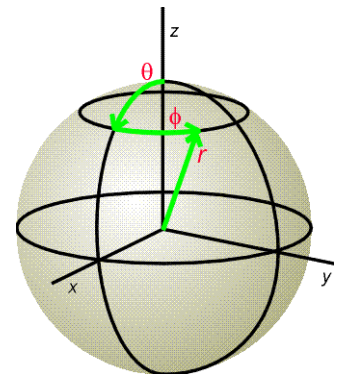
$$\lambda = \frac{h}{mv} \quad \longrightarrow \quad p = \frac{h}{\lambda} \quad \text{de Broglie's hypothesis}$$

Wavefunctions

- A particle **trajectory** is a classical concept.
- In Quantum Mechanics, a “particle” (e.g. an electron) does not follow a definite trajectory $\{\mathbf{r}(t), \mathbf{p}(t)\}$, but rather it is best described as being **distributed through space like a wave**.

Definitions

- **Wavefunction** (ψ) – a wave representing the spatial distribution of a “particle”.
- e.g. electrons in an atom are described by a wavefunction centred on the nucleus.
- ψ is a function of the coordinates defining the position of the classical particle:
 - 1-D $\psi(x)$
 - 3-D $\psi(x,y,z) = \psi(\mathbf{r}) = \psi(r,\theta,\phi)$ (e.g. atoms)
- ψ may be time dependent – e.g. $\psi(x,y,z,t)$



The Importance of ψ

- ψ completely defines the system (e.g. electron in an atom or molecule).
- If ψ is known, we can determine any observable property (e.g. energy, vibrational frequencies, ...) of the system.
- QM provides the tools to determine ψ computationally, to interpret ψ and to use ψ to determine properties of the system.

To get full knowledge of a problem **POSITION** and **MOMENTUM** need to be known

For **QUANTUM particles** its Uncertain $\rightarrow \Delta x \Delta p_x \geq \frac{\hbar}{2}$

First Postulate Introduce a parameter called

$$\Psi(x)$$

How to obtain? (We are yet to see!)

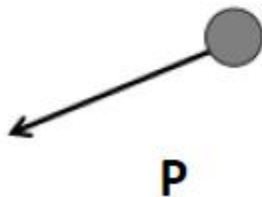
Δx and Δp_x are not absolute values $\rightarrow$ they represent the accuracy of measurement

Classical particle

Hello, my name is:

Classical particle

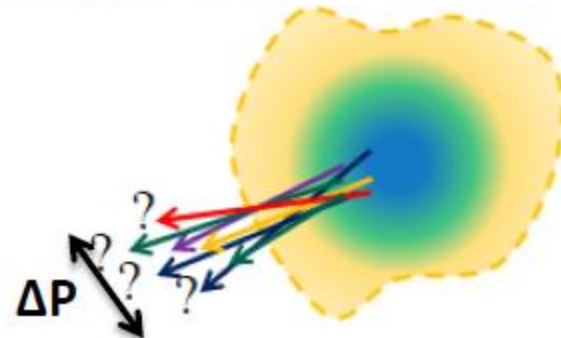
my position is 11.2392... Ang
my momentum is -23.1322... m/s



Quantum particle

Wavefunction = $\psi(x)$

"I can tell you my exact position, but then I can't tell you my momentum. I can tell you my exact momentum, but then I can't tell you my position. I can give you a pretty good estimate of my position, but then I have to give you a bad estimate of my momentum. I can..."



Quantum Mechanics says $\rightarrow$ The state of a Quantum Object is completely specified by a wavefunction $\Psi(x)$.



Does it mean something?



Probability

$\Psi^*(x)\Psi(x)$ is the probability density.

For Quantum Particles probability gives full information.

Probability that the particle can be found in the interval dx

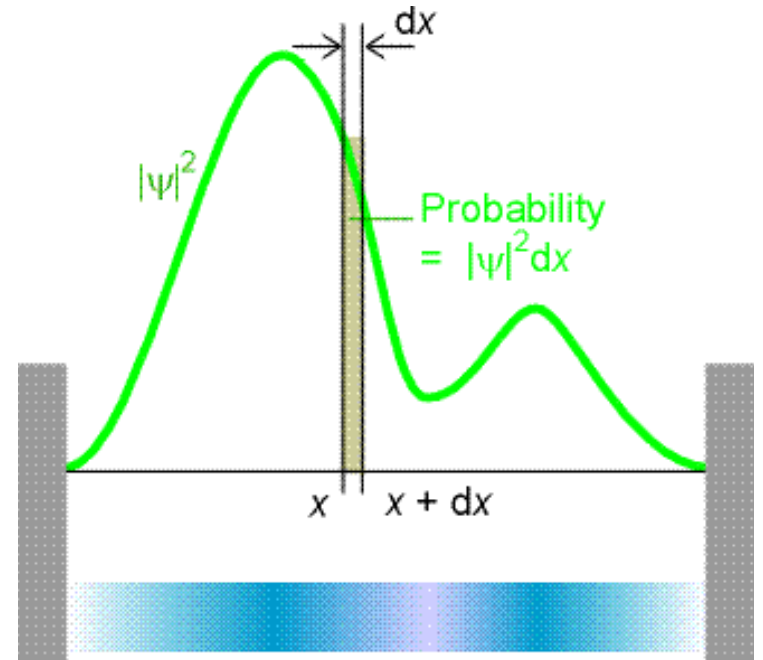
$$\Psi^*(x)\Psi(x)dx$$

A given wavefunction of the form $\Psi(x) = Af(x)$ needs to get normalized such that total Probability = 1, using the condition for **NORMALIZATION**

$$\int_{all} \Psi^*(x)\Psi(x)dx = 1$$

Suitable A is obtained

Probability must be real and positive
($0 \leq P \leq 1$).

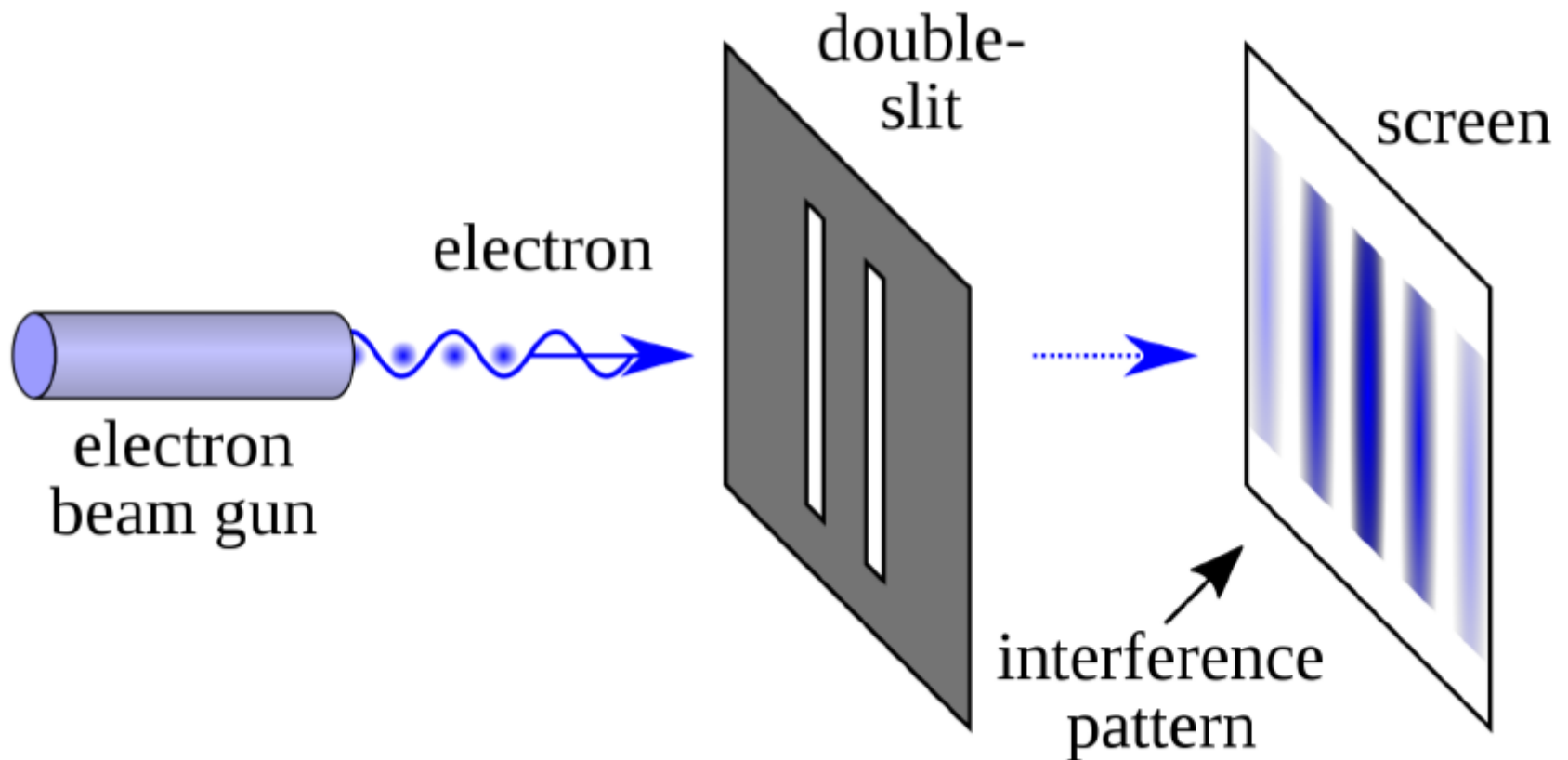


Linearity and Superposition

Let $\Psi_1(x)$ and $\Psi_2(x)$ represent the two wave functions of a quantum system, then the linear superposition will also be a valid wavefunction for the same quantum system.

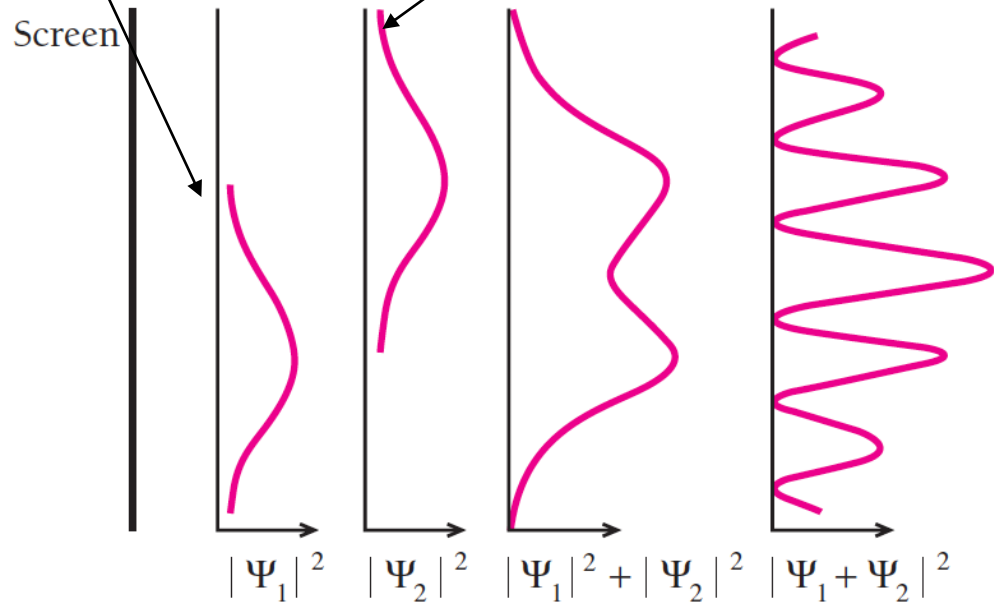
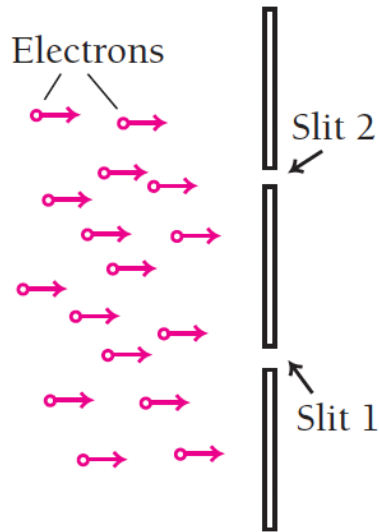
$$\Psi(x) = a_1 \Psi_1(x) + a_2 \Psi_2(x)$$

Demonstration of superposition



$$P_1 = |\Psi_1|^2 = \Psi_1^* \Psi_1$$

$$P_2 = |\Psi_2|^2 = \Psi_2^* \Psi_2$$



$$\Psi = \Psi_1 + \Psi_2$$

$$P = |\Psi|^2 = |\Psi_1 + \Psi_2|^2 = (\Psi_1^* + \Psi_2^*)(\Psi_1 + \Psi_2)$$

$$= \Psi_1^* \Psi_1 + \Psi_2^* \Psi_2 + \Psi_1^* \Psi_2 + \Psi_2^* \Psi_1$$

$$= P_1 + P_2 + \Psi_1^* \Psi_2 + \Psi_2^* \Psi_1$$

The wave functions Ψ_1 and Ψ_2 add to produce the intensity at the screen, not the probability densities $|\Psi_1|^2$ and $|\Psi_2|^2$.

Expectation value

Let $\Psi(x)$ describe the wavefunction of a quantum system, average value of the position (x) is given by the expectation value integral:

$$\langle x \rangle = \int \Psi^*(x) x \Psi(x) dx$$

Expectation value of position of a quantum particle

Probability that the particle can be found in the interval dx

$$\Psi^*(x)\Psi(x)dx$$

What is the expectation value of the position of the particle ?

$$\langle x \rangle = \frac{P_1 x_1 + P_2 x_2}{P_1 + P_2}$$

$$\langle x \rangle = \frac{\int_{all} x \Psi^*(x) \Psi(x) dx}{\int_{all} \Psi^*(x) \Psi(x) dx}$$

One-electron Schrödinger equation

General Formulation

- (1) There exist a wavefunction, $\Psi(x, y, z, t)$, describing the dynamic behavior of the system. $\searrow$ mathematically complex quantity
- (2) Ψ and $\nabla\Psi$ must be continuous, finite, and single-valued for all values of x, y, z , and t .
- (3) $\Psi^* \Psi dV = |\Psi|^2 dV$: the probability that the particle will be found in the spatial volume element dV .
- $$\int \Psi^* \Psi dV = 1: \text{normalization}$$

Wave Mechanics

- The wavefunction (ψ) contains all the information we need to know about any particular system.
- How do we determine ψ and use it to deduce properties of the system?

Operators and Observables

- If ψ is the wavefunction representing a system, we can write:

$$\hat{Q}\psi = Q\psi$$

where Q – “observable” property of system (e.g. energy, momentum, dipole moment ...)

$\hat{Q}$ – operator corresponding to observable Q .

- This is an **eigenvalue equation** and can be rewritten as:

$$\hat{Q}(\psi) = Q \times \psi$$

operator Q acting on function ψ (eigenfunction)	function ψ multiplied by a number Q (eigenvalue)
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(**Note:** ψ can't be cancelled).

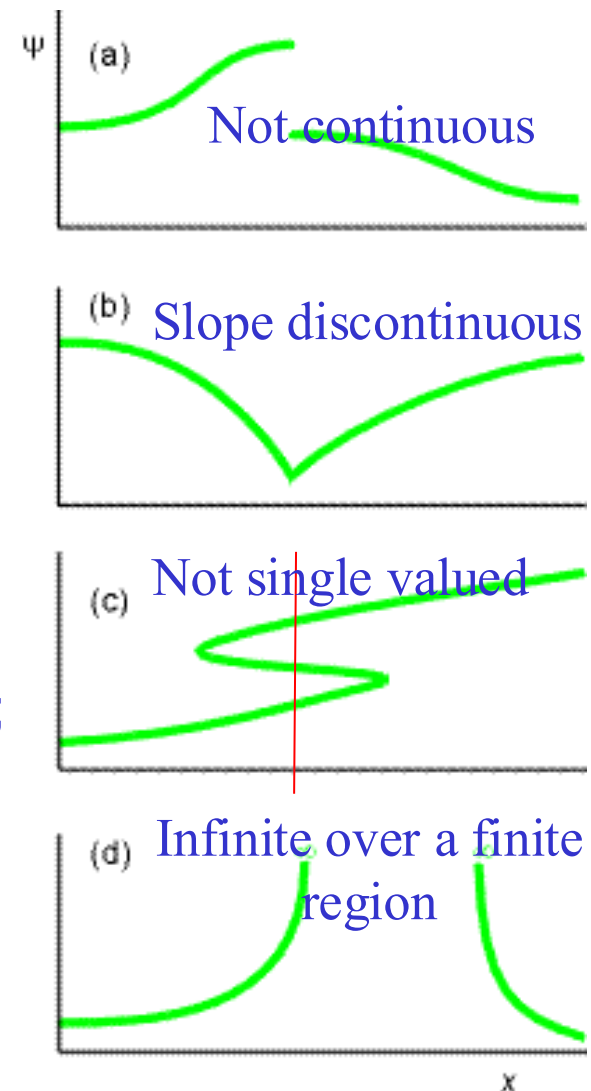
Examples: $d/dx (e^{ax}) = a e^{ax}$

$$d^2/dx^2 (\sin ax) = -a^2 \sin ax$$

The Wavefunction

The Born interpretation of ψ places restrictions on the form of the wavefunction:

- (a) ψ must be continuous (no breaks);
- (b) The gradient of ψ ($d\psi/dx$) must be continuous (no kinks);
- (c) ψ must have a single value at any point in space;
- (d) ψ must be finite everywhere;
- (e) ψ cannot be zero everywhere.



- Other restrictions (**boundary conditions**) depend on the exact system.
- These restrictions on ψ mean that only certain wavefunctions and $\therefore$ only certain energies of the system are allowed.

$\therefore$ Quantization of $\psi \Rightarrow$ Quantization of E

To find ψ and calculate the properties (observables) of a system:

1. Construct relevant operator $\hat{Q}$
2. Set up equation $\hat{Q}\psi = Q\psi$
3. Solve equation $\Rightarrow$ allowed values of ψ and Q .

Quantum Mechanical Position and Momentum Operators

1. Operator for position in the x-direction is just multiplication by x

$$\hat{x}\psi = x\psi$$

2. Operator for linear momentum in the x-direction: $\hat{p}_x = \left(\frac{\hbar}{i}\right) \left(\frac{d}{dx}\right)$

$$\hat{p}_x\psi = p_x\psi \Rightarrow \frac{\hbar}{i} \cdot \frac{d\psi}{dx} = p_x\psi$$

(solve first order differential equation $\Rightarrow \psi$, p_x).

Physical properties		Operators	
Name of Operator	Observables	Operators	Symbols
Position	Position with x coordinate	x	$\hat{x}$
Momentum	x component of momentum	$-i\hbar \frac{\partial}{\partial x}$	$\widehat{p_x}$
Kinetic Energy	Kinetic Energy	$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$	$\hat{T}$
Potential Energy	Potential Energy	V(x)	$\hat{V}$
Total Energy (E)	Hamiltonian Operator (Time-independent)	$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$	$\hat{H}$
Total Energy (E)	Hamiltonian Operator (Time-dependent)	$-i\hbar \frac{\partial}{\partial t}$	$\hat{H}$

The Schrödinger Equation (1926)

- The central equation in Quantum Mechanics.
- Observable = total energy of system.

Schrödinger Equation $\hat{H}\psi = E\psi$ $\hat{H}$ Hamiltonian Operator
E Total Energy

where $\hat{H} = \hat{T} + \hat{V}$ and $E = T + V$.

- SE can be set up for any physical system.
- The form of $\hat{H}$ depends on the system.
- Solve SE $\Rightarrow \psi$ and corresponding E .

Examples

1. Particle Moving in 1-D $\psi(x)$

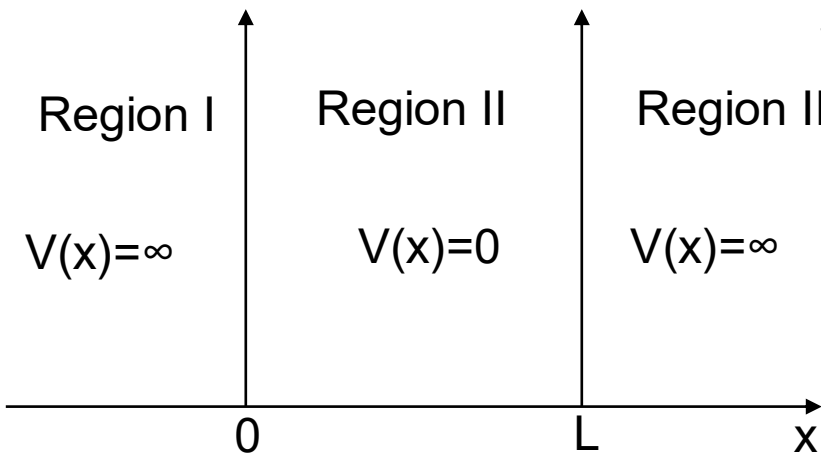
$$\hat{H}\psi = \hat{T}\psi + \hat{V}\psi = E\psi \longrightarrow -\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}\right) + V(x)\psi = E\psi$$

- The form of $V(x)$ depends on the physical situation:
 - Finite well $V(x) = 0$ for all a range of x and finite for rest.
 - Infinite well $V(x) = 0$ for all a range of x and infinite for rest.
 - Harmonic oscillator $V(x) = \frac{1}{2}kx^2$

The Particle in A Box

Particle in 1-dimensional box

- Infinite walls



Time Independent Schrödinger Equation

$$\underbrace{\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2}}_{\text{KE}} + \underbrace{V(x)\psi}_{\text{PE}} = \underbrace{E\psi}_{\text{TE}}$$

System

Particle of mass **m** in 1-D box of length **L**.

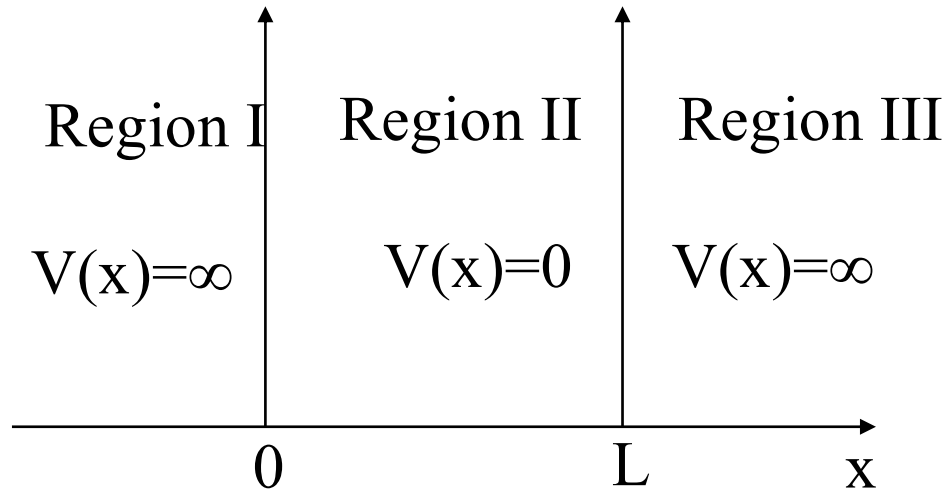
Position **x = 0 → L**.

Particle cannot escape from box as PE **V(x) = ∞** for **x = 0, L** (walls).
 PE inside box: **V(x) = 0** for **0 < x < L**.

Restrictions on ψ

- In principle **Schrödinger Eqn.** has an infinite number of solutions.
- So far we have general solutions: any value of $\{\psi, E\}$.
- **BUT** – due to the **Born interpretation of ψ** , only certain values of ψ are physically acceptable:
 - outside box ($x < 0, x > L$) $V = \infty \Rightarrow$ impossible for particle to be outside the box
 $\therefore |\psi|^2 = 0 \Rightarrow \psi = 0$ outside box.
 - ψ must be a continuous function
 $\Rightarrow$ **Boundary Conditions** $\psi = 0$ at $x = 0$
 $\psi = 0$ at $x = L$.

Applying boundary conditions: Restrictions on ψ



Region I and III:
$$\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \infty * \psi = E\psi \quad \Rightarrow |\psi|^2 = 0$$

Region II:

$$\frac{d^2\Psi(x)}{dx^2} = -\frac{2mE}{\hbar^2}\Psi(x)$$

Second derivative of a function equals a negative constant times the same function.

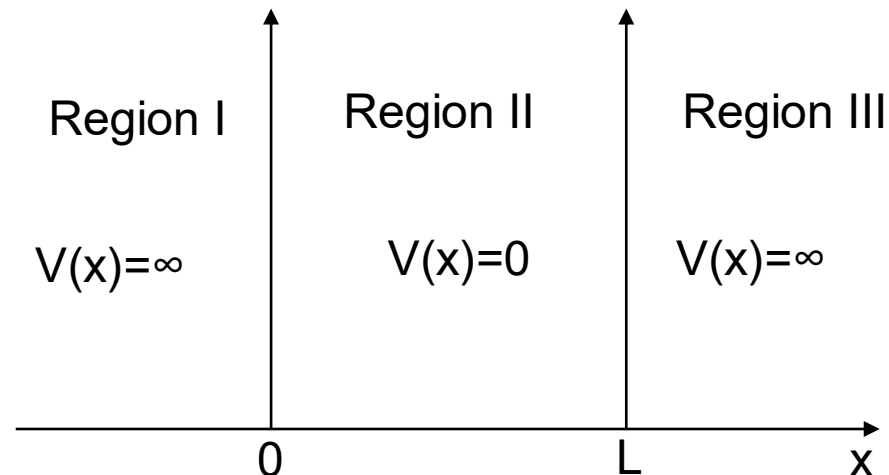
$$\frac{d^2\Psi(x)}{dx^2} = -k^2\Psi(x)$$

$$k^2 = \frac{2mE}{\hbar^2}$$

Functions with this property $\longrightarrow$ **sin and cos**

$$\frac{d^2 \sin(ax)}{dx^2} = -a^2 \sin(ax)$$

$$\frac{d^2 \cos(ax)}{dx^2} = -a^2 \cos(ax)$$



Region II:

$$\frac{d^2\Psi(x)}{dx^2} = -\frac{2mE}{\hbar^2}\Psi(x)$$

$$\frac{d^2\Psi(x)}{dx^2} = -k^2\Psi(x) \quad k^2 = \frac{2mE}{\hbar^2}$$

This is similar to the general differential equation:

$$\psi = A \sin kx + B \cos kx$$

Applying boundary conditions:

$$\text{At } x = 0 \rightarrow \psi = 0$$

$$0 = A \sin 0k + B \cos 0k$$

$$0 = 0 + B * 1$$

$$\Rightarrow B = 0$$

$$\text{At } x = L \psi = 0$$

$$0 = A \sin kL$$

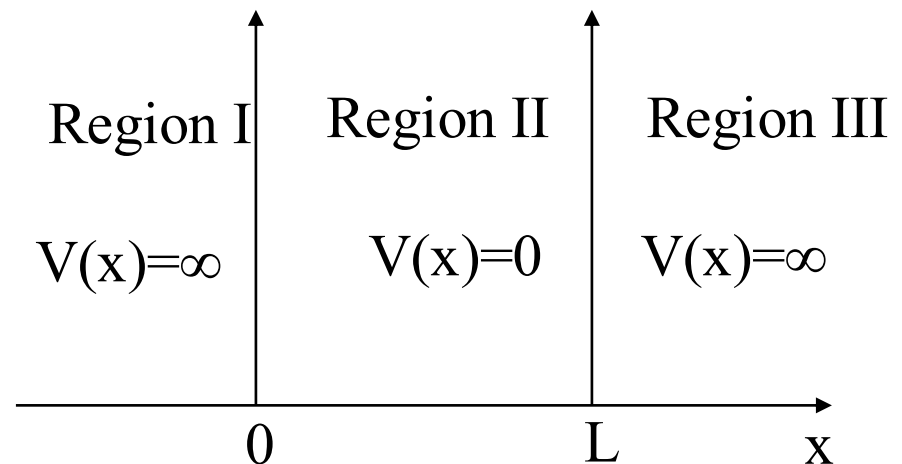
$$\text{But } A \neq 0$$

$$kL = n\pi$$

Thus, wave function:

$$\psi_{II} = A \sin \frac{n\pi x}{L}$$

But what is 'A'?



Normalizing wave function:

$$\int_0^L (A \sin kx)^2 dx = 1$$

$$\sin^2 a = \frac{(1 - \cos(2a))}{2}$$

$$\int_0^L A^2 \left(\frac{1}{2} - \frac{\cos 2kx}{2} \right) dx = 1$$

$$|A|^2 \left[\frac{x}{2} - \frac{\sin 2kx}{4k} \right] \Big|_0^L = 1$$

$$|A|^2 \left[\frac{L}{2} - \frac{\sin 2 \frac{n\pi}{L} L}{4 \frac{n\pi}{L}} \right] = 1$$

$$k = \frac{n\pi}{L}$$

$$\sin 2\pi = 0$$

$$|A|^2 \left(\frac{L}{2} \right) = 1$$

$$|A| = \sqrt{\frac{2}{L}}$$

Thus normalized wave function is:

$$\psi_{II} = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

Calculating Energy Levels:

$$k^2 = \frac{2mE}{\hbar^2}$$

$$E = \frac{k^2 \hbar^2}{2m}$$

$$E = \frac{k^2 h^2}{2m4\pi^2}$$

$$k = \frac{n\pi}{L}$$

$$(\hbar = \frac{h}{2\pi})$$

$$E = \frac{n^2 \pi^2}{L^2} \frac{h^2}{2m4\pi^2}$$

Thus Allowed Energies, E_n , are:

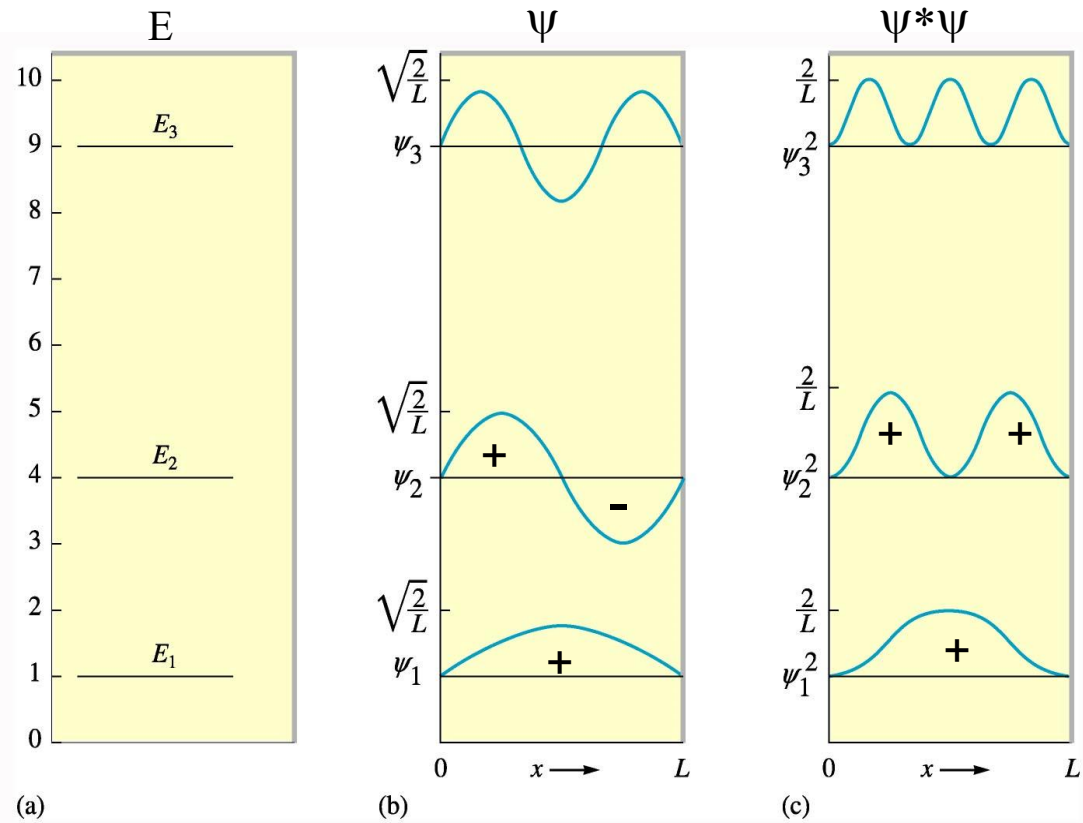
$$E_n = \frac{n^2 h^2}{8mL^2}$$

Particle in a 1-Dimensional Box

$$E_n = \frac{n^2 h^2}{8mL^2}$$

$$\psi_{II} = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

$$|\psi_{II}|^2 = \frac{2}{L} \left(\sin \frac{n\pi x}{L} \right)^2$$



Difference b/w adjacent energy levels:

$$\Delta E = E_{n+1} - E_n = (2n + 1)E_1$$